

SECTION 1 (13 marks)

- (a) Express $z_1 = 2 + 2i$ and $z_2 = -\sqrt{3} + i$ in mod arg form and use to evaluate $\frac{z_1^3}{z_2^2}$ [3]
(leave answer in mod arg form)

- (b) If $z = x + iy$ show $z + \frac{|z|^2}{z} = 2 \operatorname{Re} z$ [3]

- (c) Sketch the region in the Argand diagram defined by

i) $\frac{\pi}{3} \leq \operatorname{Arg} z \leq \frac{2\pi}{3}$ [2]

ii) $|z - i| = |z|$ [2]

- (d) Solve for z (giving answer in the form $a + bi$.) $z(1 + i) = 2 + 6i$ [3]

QUESTION 2 (13 marks)

- (a) If $z = 2 - i$. Find the real numbers p and q such that $pz + \frac{q}{z} = 1$ [3]

- (b) i) Find the roots of $z^5 + 1 = 0$ and show their position on a unit circle in an Argand diagram. [3]

ii) Since the sum of these roots add to zero show $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$ [2]

- (c) The locus of a point $P(x, y)$, which moves in the complex plane is represented by the equation $|z - 3i| = 2$

- i) Draw this locus, give a geometrical description and hence write down its Cartesian equation [2]

- ii) Show that the minimum value of $\arg z$ is $\cos^{-1}\left(\frac{2}{3}\right)$ and find the modulus of z when P is in the position of minimum argument. [3]

QUESTION 3 (15 marks)

- (a) i) Using the fact that if $z = \cos \theta + i \sin \theta$, then $z^n + z^{-n} = 2 \cos n\theta$, show

$$\cos^3 \theta = \frac{1}{4}(\cos 3\theta + 3 \cos \theta) \quad [2]$$

ii) Use the above to evaluate $\int_0^{\frac{\pi}{3}} \cos^3 \theta \, d\theta$

- (b)) If $z=a+bi$ where $a^2 + b^2 \neq 0$

i) Show that if $\text{Im}(z)>0$ then $\text{Im}\left(\frac{1}{z}\right)<0$. [2]

ii) Prove that $\left|\frac{1}{z}\right| = \frac{1}{|z|}$ [2]

- (c) z_1 and z_2 are complex numbers such that $|z_1|=1$ and $|z_2|=4$ and $\arg \frac{z_2}{z_1} = \frac{2\pi}{3}$

i) Show $|\sqrt{z_1} - \sqrt{z_2}| = \sqrt{3}$ [3]

ii) If $\arg \sqrt{z_1} = \alpha$ and $(0 < \alpha < \frac{\pi}{2})$ find $\arg (\sqrt{z_1} - \sqrt{z_2})$ in terms of α [2]

iii) Explain, with the aid of a diagram, why there are two solutions to part (ii)
if $0 \leq \alpha \leq 2\pi$ [1]

Question 1.

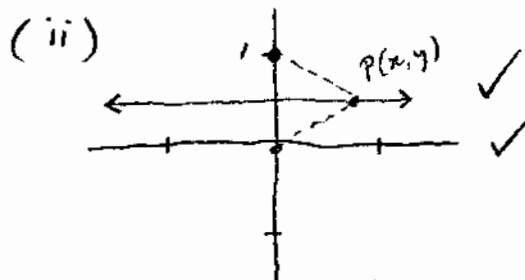
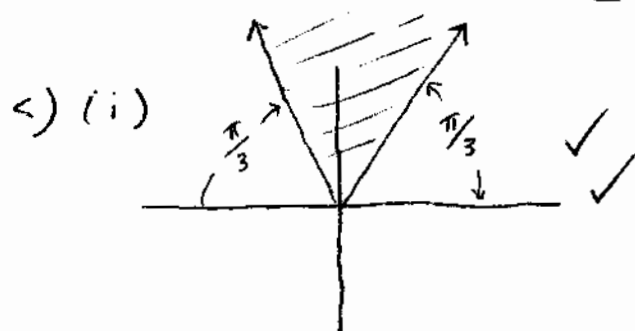
a) $z_1 = 2\sqrt{2} \operatorname{cis} \frac{\pi}{4}$, $z_2 = 2 \operatorname{cis} \frac{5\pi}{6}$ ✓ [3]

$$\frac{z_1^3}{z_2^2} = \frac{(2\sqrt{2})^3 \operatorname{cis} \frac{3\pi}{4}}{2^2 \operatorname{cis} \frac{10\pi}{6}} = 2\sqrt{2} \operatorname{cis} \left(\frac{9\pi}{12} - \frac{20\pi}{12} \right) = 2\sqrt{2} \operatorname{cis} \frac{-11\pi}{12} \checkmark \checkmark$$

b) $z + \frac{|z|^2}{z} = x + yi + \frac{x^2 + y^2}{x + iy} = \frac{(x + iy)^2 + x^2 + y^2}{x + iy}$ ✓ [3]

$$= \frac{x^2 + 2ixy - y^2 + x^2 + y^2}{x + iy} = \frac{2x^2 + 2ixy}{x + iy} = \frac{2x(x + iy)}{x + iy} = 2x \checkmark$$

Since $\operatorname{Re} z = x$, $z + \frac{|z|^2}{z} = 2\operatorname{Re} z \checkmark$



d) $z = x + yi$ [3]

$$(x + yi)(1 + i) = x + xi + yi - y = (x - y) + i(x + y)$$

$$\therefore (x - y) + i(x + y) = 2 + 6i \checkmark$$

$$x - y = 2$$

$$x + y = 6$$

$$\therefore 2x = 8, \quad x = 4, \quad y = 2 \checkmark$$

$$\therefore z = 4 + 2i \checkmark$$

Question 2.

$$a) p(2-i) + \frac{q}{2-i} \times \frac{2+i}{2+i} = 2p - ip + \frac{2q + iq}{4+1} \quad [3]$$

$$= \frac{10p - 5pi + 2q + iq}{5} = 1 \quad \therefore 10p + 2q + i(-5p + q) = 5$$

$$\therefore 10p + 2q = 5 \text{ and } -5p + q = 0$$

$$\begin{array}{rcl} 10p + 2q & = & 5 \\ -10p + 2q & = & 0 \\ \hline 4q & = & 5, \quad q = \frac{5}{4} \quad \checkmark \\ p & = & \frac{1}{4} \end{array}$$

[3]

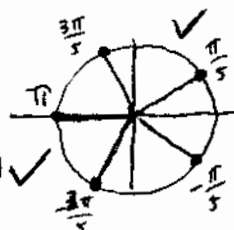
$$b) i) z^5 = -1 \text{ if } z = r(\cos \theta + i \sin \theta), \quad z^5 = r^5(\cos \theta + i \sin \theta)^5$$

$$= r^5(\cos 5\theta + i \sin 5\theta) = -1, \text{ since } |-1| = 1 \quad r^5 = 1, \quad r = 1$$

$$\therefore \cos 5\theta + i \sin 5\theta = -1 \quad \therefore \cos 5\theta = -1, \quad 5\theta = \pi + 2k\pi \quad \checkmark$$

$$\theta = \frac{\pi + 2k\pi}{5} = \frac{\pi}{5}, \frac{3\pi}{5}, -\frac{\pi}{5}, \pi, -\frac{3\pi}{5}$$

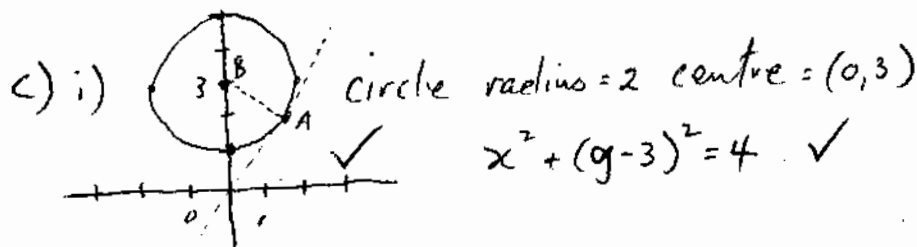
$$\therefore \text{Roots are } -1, \text{cis } \frac{\pi}{5}, \text{cis } (-\frac{\pi}{5}), \text{cis } (\frac{3\pi}{5}), \text{cis } (-\frac{3\pi}{5}) \quad \checkmark$$



ii) Sum of roots are

[2]

$$\begin{aligned} & -1 + \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} + \cos(-\frac{\pi}{5}) + i \sin(-\frac{\pi}{5}) + \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} + \cos(-\frac{3\pi}{5}) + i \sin(-\frac{3\pi}{5}) \\ & = -1 + \cos \frac{\pi}{5} + i \sin \frac{\pi}{5} + \cos \frac{\pi}{5} - i \sin \frac{\pi}{5} + \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} + \cos \frac{3\pi}{5} - i \sin \frac{3\pi}{5} = 0 \\ & = -1 + 2\cos \frac{\pi}{5} + 2\cos \frac{3\pi}{5} = 0 \quad \therefore \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2} \quad \checkmark \end{aligned}$$



[2]

$$x^2 + (y-3)^2 = 4 \quad \checkmark$$

[3]

ii) Minimum arg when Z is tangent OA $\checkmark$

$\angle COA = \angle OBA$ (both $90^\circ - \angle AOB$)

$\angle OBA = \cos^{-1} \frac{2}{3} \quad \therefore \angle COA = \arg Z = \cos^{-1} \frac{2}{3} \quad \checkmark$

$|Z| = \sqrt{5} \quad \checkmark$

ion 3

$$(2 \cos \theta)^3 = (z' + z^{-1})^3 = z^3 + 3z + 3z^{-1} + z^{-3} \quad [2]$$

$$8 \cos^3 \theta = (z^3 + z^{-3}) + 3(z + z^{-1}) = 2 \cos 3\theta + 6 \cos \theta \quad \checkmark$$

$$\therefore \cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta = \frac{1}{4} (\cos 3\theta + 3 \cos \theta) \quad \checkmark$$

$$\text{ii) } \frac{1}{4} \int_0^{\frac{\pi}{3}} \cos 3\theta + 3 \cos \theta d\theta = \frac{1}{4} \left[\frac{1}{3} \sin 3\theta + 3 \sin \theta \right]_0^{\frac{\pi}{3}} \quad [3]$$

$$= \frac{1}{4} \left[(0 + 3 \sin \frac{\pi}{3}) - (0 + 0) \right] = \frac{1}{4} \left(\frac{3\sqrt{3}}{2} \right) = \frac{3\sqrt{3}}{8} \quad \checkmark$$

- 1/2 for computational error.

b) (i) If $\text{Im}(z) > 0$ and $z = a + bi$ then $b > 0$ [2]

$$\frac{1}{z} = \frac{1}{a+bi} \times \frac{a-bi}{a-bi} = \frac{a-bi}{a^2+b^2} \quad \checkmark$$

$$\therefore \text{Im}\left(\frac{1}{z}\right) = \frac{-b}{a^2+b^2} \text{ since } b > 0 \text{ and } a^2+b^2 > 0$$

$$\text{Im}\left(\frac{1}{z}\right) < 0 \quad \checkmark$$

No penalty for not stating $a^2+b^2 > 0$

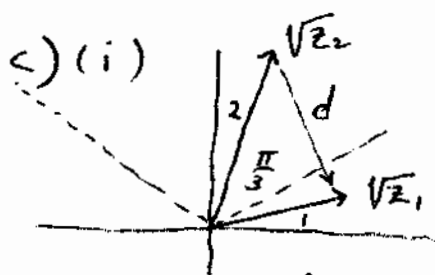
No penalty for

$$\text{Im}\left(\frac{a+bi}{a^2+b^2}\right) = \frac{-bi}{a^2+b^2}$$

(ii) If $z = r \text{cis } \theta$, $|z| = r$ and $\frac{1}{|z|} = \frac{1}{r}$ [2]

$$\frac{1}{z} = \frac{1}{r} \text{cis}(-\theta), \quad \left|\frac{1}{z}\right| = \frac{1}{r} \therefore \left|\frac{1}{z}\right| = \frac{1}{|z|} \quad \checkmark$$

- 1/2 for minor transcription errors



$$\arg \frac{z_2}{z_1} = \arg z_2 - \arg z_1 = \frac{2\pi}{3} \quad [3]$$

$$\arg \sqrt{z_2} = \frac{1}{2} \arg z_2, \arg \sqrt{z_1} = \frac{1}{2} \arg z_1 \quad \checkmark$$

$$\therefore \arg \sqrt{z_2} - \arg \sqrt{z_1} = \frac{1}{2} (\arg z_2 - \arg z_1) = \frac{\pi}{3}$$

Using cosine rule $d^2 = 2^2 + 1^2 - 2(2)(1) \cos \frac{\pi}{3} = 5 - 4 \cos \frac{\pi}{3} = 3 \quad \checkmark$

$$\therefore |\sqrt{z_1} - \sqrt{z_2}| = \sqrt{3} \quad \checkmark$$

i) By Pythagoras $\angle OAB = \frac{\pi}{2} \therefore \angle OBA = \frac{\pi}{6} \quad \checkmark$ [2]

$$\angle OBC = \frac{\pi}{3} + \alpha \therefore \beta = \pi - \left(\frac{\pi}{3} + \alpha + \frac{\pi}{6}\right) = \frac{\pi}{2} - \alpha \quad \checkmark$$

Since $\angle DBA$ is principal argument of $(\sqrt{z_1} - \sqrt{z_2})$

$$\arg(\sqrt{z_1} - \sqrt{z_2}) = \alpha - \frac{\pi}{2} \quad \checkmark$$

i) AB would be other answer [1]